

DEVELOP SOFTWARE USING ARTIFICIAL INTELLIGENCE (AI) TOOLS / AI AGENTS FOR THE FOLLOWING USER STORY

User Story

Customer Segmentation & Personalized Offer Generator Platform

As a regional/market manager at Communication Service Provider, I want a simple customer segmentation and personalized offer generator tool that helps me classify customers based on their profile, value, behavioural data, usage, lifecycle and current contract details so that I can see the customer segmentation and tailored personalized offer for the customer. From the list of offers/promotions, map the offer to the right customer segment without needing advanced analytics.

Objective

To develop a basic software module that enables Regional/Market Manager team to segment customers and view tailored personalized offers using a simple value-based analysis accordingly.

Steps

- **Collect Customer Data:** Gather customer profile information (demographics, preferences, location, Hobby etc.) and collect behavioural data (purchase history, service usage, feedback, interaction logs).
- **Customer Segmentation Logic:** Segmentation based on specific customer criteria:
 - **Usage-Based Segmentation:** It is based on customer usage patterns. This includes data usage, call duration, text message volume, and app usage. This segmentation helps tailor service plans, ensuring customers have the right amount of data or minutes to fit their needs.
 - **Customer Value Segmentation:** Customers can be divided based on their lifetime value to CSP. High-value customers who consistently subscribe to premium plans may receive exclusive offers or personalized customer support, while cost-conscious customers may be offered more budget-friendly options.
 - **Bundling Behaviour Segmentation:** It is based on customer preferences for bundled services. Some customers prefer all-in-one packages that include internet, TV, and phone services, while others may only want one specific service. Telecom companies can segment and offer tailored bundle options.
 - **Contract Length Segmentation:** Segmenting customers by their contract lengths (e.g., month-to-month, annual contracts) allows CSPs to customize retention strategies. Customers nearing

the end of their contracts may receive renewal incentives, while those on month-to-month plans may receive special offers to encourage longer commitments.

- **Map Offers to Segments:** Review the list of available offers/promotions, define criteria for each offer (eligibility, usage thresholds, customer preferences), and match each segment to the most relevant offers.
- **Generate Personalized Offers:** Select the best-fit offer/promotion for each customer and personalize the offer message using customer-specific details (e.g., name, recent activity, preferences).
- **Deliver Offers:** Communicate the personalized offer to the customer through their preferred channel (email, SMS, app notification, etc.).

Data Requirements

- List of sample telecom offers and customer profiles including past feedback and interactions.
- Customer profile data (demographics, preferences, location).
- Behavioural and usage data (purchase/service history, feedback, interaction logs).
- List of available offers/promotions with eligibility criteria.
- (Optional) Historical data on offer effectiveness for further optimization.

Expected Output

A GenAI agent interface that provides:

- Personalized offer or promotion tailored to profile and behaviour for each customer.
- Improved customer engagement and higher conversion rates for offers.
- Reports or dashboards summarizing offer distribution, acceptance, and effectiveness by segment.

Key Features

- **Login Screen:** 'Administrator' to set rules; 'Regional Manager' to see Dashboard and export data.
- **Rule Engine:** Configure rules for classifying customers into different segments.
- **Automatic Classification:** System assigns tier based on defined rules.
- **Dashboard:** Displays segmented customers, their profiles, and values with filters and summary totals. Personalized offer generation status and summary of offers generated.
- **Export Option:** Download customer segmentation data as CSV or PDF for offline use.
- Tailored offer generation and performance summary dashboards.

Deliverables

All deliverables are expected to be of production-level quality, ensuring excellence in both content and look & feel:

1. Functional requirements document
2. Application Architecture and Architecture document
3. Database design / Data Models - No JSON input of data, use any RDBMS; deliverables include logical model, physical model, and ER diagram
4. UI/UX/Wireframe Design
5. Software code & Working Software
6. Functional Test scripts in CSV or Excel
7. Traceability Matrix in CSV or Excel
8. End User manual - Screens and help guide
9. AI Agent - 'Code Quality Checker' built with an AI framework and Streamlit or Gradio; takes code and checklist as input and provides observations against each checklist item and suggestions for improvements if any

Instructions

- **Use AI Tools / BYOAA:** Use AI tools or Build Your Own AI Agents (BYOAA) to develop the software and deliverables.
- **Selection of Tools:** Carefully evaluate and select the most suitable AI tools or frameworks for each phase of the project. If building custom AI agents, ensure they are scalable and maintainable. Additional weightage will be given if team builds their own AI agent to create the deliverables.
- **Quality Standards:** Adhere strictly to coding standards and best practices for software development. Apply consistent design patterns, code reviews, and automated testing to maintain high quality.
- **Integration:** Ensure seamless integration of components with the overall application architecture.
- **Testing:** Prepare comprehensive test cases, including both functional and non-functional scenarios.
- **Documentation:** Maintain clear and detailed documentation for all deliverables, including architecture diagrams, data models, and user manuals. Update documents regularly as changes are made.
- **User Experience:** Prioritise a user-centric approach in UI/UX design. Avoid using Streamlit or Gradio for the main user interface.